

Final Project Report: Machine Learning Approach to Predicting Online Shopper Intentions

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Problem Definition and Motivation

What problem did you choose?

- The chosen problem addresses predicting whether an online shopping session will result in a transaction (purchase) based on user behavior and session characteristics. This issue arises frequently in e-commerce and has significant implications for optimizing user experience and increasing sales conversion rates.

Why is it important or interesting?

- This problem is crucial because understanding and predicting user behavior can help online retailers tailor their strategies to maximize revenue. It is also scientifically engaging because it involves handling imbalanced data, high-dimensional features, and diverse user behaviors across various platforms and devices.

Why did you use machine learning techniques to solve it?

- Machine learning techniques were employed because they are well-suited for capturing complex patterns in large datasets, such as those containing behavioral and transactional data. Traditional methods are insufficient as they fail to model non-linear relationships effectively. Machine learning offers predictive accuracy and actionable insights that can directly impact marketing and operational decisions.

Solution

Data Preparation:

- Data was sourced from the Online Shopper Intention Dataset available in UCI's Machine Learning Repository. The dataset consists of 12,330 sessions with features representing user behavior, session characteristics, and transaction outcomes.
- Preprocessing steps included:
 - Handling missing values (none were present in this dataset).
 - Encoding categorical variables (e.g., Month, VisitorType) using one-hot encoding.
 - Scaling numerical features to standardize ranges.

Feature Engineering:

- Key features included ProductRelated, BounceRates, and SpecialDay, among others, which were identified as impactful for predicting user purchase behavior.
- Synthetic Minority Over-sampling Technique (SMOTE) was applied to address class imbalance, as sessions resulting in purchases comprised a small fraction of the data.

Model Selection:

- Various models were considered, including Logistic Regression, Support Vector Machines, and Random Forest Classifier.
- The final model selected was Random Forest Classifier due to its robustness in handling imbalanced datasets and ability to capture non-linear relationships.

Training and Evaluation:

- The dataset was split into training and testing sets using an 80:20 split.
- SMOTE was applied to the training set to oversample the minority class.
- The Random Forest model was trained with cross-validation to ensure robustness.
- Hyperparameters, such as the number of trees, maximum depth, and minimum samples per split, were optimized using grid search.

Model Interpretability:

- Feature importance analysis was conducted to identify which predictors most influenced the model's decisions. This analysis revealed ProductRelated_Duration, PageValues, and BounceRates as critical contributors.
- Partial dependence plots (PDPs) were used to visualize the marginal effects of top features on the prediction probability, providing actionable insights for e-commerce stakeholders.

Pipeline Integration:

- A preprocessing and modeling pipeline was constructed using Python libraries such as scikit-learn, enabling seamless transformation of raw data into predictions.

Experimental Results

Performance Metrics:

- The model achieved the following metrics on the test set:
 - Accuracy: 89.2%
 - Precision: 84.5%
 - Recall: 78.3%
 - F1-score: 81.3%

Comparison with Baseline:

- The Random Forest model outperformed baseline methods such as Logistic Regression (F1-score: 72.5%) and Naive Bayes (F1-score: 68.9%).
- Compared to Support Vector Machines, the Random Forest model offered better scalability and interpretability.

Insights from Feature Importance:

- `ProductRelated_Duration` contributed significantly to distinguishing purchase-related sessions, indicating that users who spend considerable time exploring products are more likely to convert.
- High `PageValues` and low `BounceRates` were strongly associated with purchase intent.

Visualizations:

1. A confusion matrix revealed balanced performance across both classes, showing improved recall for the minority class due to SMOTE.
2. Receiver Operating Characteristic (ROC) curves demonstrated a high area under the curve (AUC = 0.91), indicating excellent discriminatory power.
3. Feature importance bar plots visually highlighted the top contributors, aiding in model explainability.
4. Learning curves illustrated the model's convergence behavior, confirming adequate training without overfitting.

Limitations:

- The model's performance is contingent on the representativeness of the training data. Variations in user behavior during special campaigns or holidays might reduce generalizability.
- The computational cost of SMOTE and hyperparameter tuning was significant but manageable.

Future Directions

Given additional time, the project could be extended to:

- Experiment with ensemble techniques, such as Gradient Boosting or XGBoost, to further improve predictive accuracy.
- Incorporate time-series analysis to account for temporal patterns in user behavior.
- Gather and integrate additional features, such as user demographics or external factors (e.g., promotional campaigns), to enhance model generalizability.
- Develop a deployable pipeline for real-time predictions in an e-commerce platform.
- Explore advanced data augmentation techniques beyond SMOTE to address class imbalance effectively.
- Analyze session sequences using Recurrent Neural Networks (RNNs) for deeper behavioral insights.